

# Yue Wu

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## Education

- **Ph.D. Candidate in Applied Mathematics** 09/2023–present  
Division of Applied Mathematics, Brown University, Providence, RI 02912, USA  
Advisor: *Prof. Chi-Wang Shu*
- **M.Sc. in Applied Mathematics** 09/2023–05/2025  
Division of Applied Mathematics, Brown University, Providence, RI 02912, USA
- **B.Sc. in Information & Computational Science** 09/2019–06/2023  
School of the Gifted Young, University of Science and Technology of China, Hefei, Anhui 230026, China
- Wuxi No. 1 High School, Wuxi, Jiangsu 214031, China 09/2017–06/2019

## Research Interests

- High order numerical methods for partial differential equations
  - Discontinuous Galerkin finite element methods
  - Finite difference Weighted Essentially Non-Oscillatory (WENO) methods
- Scientific computing
  - Parallel PDE solver development

## Publications and Preprints

1. **Y. Wu** and C.-W. Shu, A positivity preserving and entropy stable nodal discontinuous Galerkin scheme for ideal MHD equations, submitted to *Commun. Math. Stat.* doi:10.48550/arXiv.2604.23885.
2. **Y. Wu** and C.-W. Shu, Finite difference alternative WENO schemes with Riemann invariant-based local characteristic decompositions for compressible Euler equations, *J. Comput. Phys.* 537 (2025) 114104. doi:10.1016/j.jcp.2025.114104. MR4912873.
3. **Y. Wu** and Y. Xu, A high-order local discontinuous Galerkin method for the  $p$ -Laplace equation, *Beijing J. of Pure and Appl. Math.* 2 (1) (2025) 373–422. doi:10.4310/BPAM.250415002006.

## Research Experience

1. **Structure-preserving discontinuous Galerkin schemes for MHD equations**  
Brown University 03/2025–04/2026  
Supervisor: *Prof. Chi-Wang Shu*
  - Studied all types of explicit structure-preserving conservative schemes for compressible MHD.
  - Developed a nodal discontinuous Galerkin scheme that combines positivity preservation and entropy stability.
2. **Efficient alternative WENO (A-WENO) methods for compressible Euler equations**  
Brown University 09/2024–02/2025  
Supervisor: *Prof. Chi-Wang Shu*
  - Investigated the effect of different transform variables in the local characteristic decomposition on the performance of A-WENO methods.
  - Developed an A-WENO code using Riemann invariants as transform variables to save cost.

### 3. Discontinuous Galerkin Methods for the $p$ -Laplace Equation

Bachelor's thesis at USTC

12/2022–06/2023

Supervisor: *Prof. Yan Xu*

- Proved an a priori error estimate for an LDG scheme for the  $p$ -Laplace equation.
- Developed and implemented an efficient preconditioned gradient descent method.

### 4. Positivity-Preserving Conservative Low Rank Methods for Vlasov Dynamics

Purdue University (remote)

06/2022–08/2022

Supervisor: *Prof. Xiangxiong Zhang*

- Developed a low-rank correction algorithm with positivity preservation and orthogonality constraints via optimization, which can post-process data from a dynamic low-rank solver.

### 5. Numerical Simulation of Plasma Equilibrium Evolution in Nuclear Fusion

USTC undergraduate research project

06/2021–05/2022

Supervisor: *Prof. Mengping Zhang*

- Developed a parallel hybrid finite difference-pseudo spectral code for resistive MHD in toroidal geometry, and performed long-time simulation of resistive tearing mode instability in tokamaks.
- Checked the results with researchers from the Institute of Plasma Physics, CAS, and against those from existing open-source codes.

## Teaching Experience

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|----------------------------------------------------------------------------|-------------|
| 1. TA: APMA 1655 – <i>Intro. to Prob. &amp; Stats. with Theory</i> , Brown | Spring 2026 |
| 2. TA: APMA 0160 – <i>Intro. to Sci. Computing</i> , Brown                 | Fall 2025   |
| 3. TA: APMA 1650 – <i>Stat. Inference I</i> , Brown                        | Spring 2025 |
| 4. TA: APMA 1210 – <i>OR: Deterministic Models</i> , Brown                 | Fall 2024   |
| 5. TA: <i>Computational Methods B</i> , USTC                               | Spring 2022 |

## Presentations and Workshops

1. Poster session, the 2024 International Congress of Basic Science (ICBS), Beijing, China 07/2024

## Professional Services

1. Reviewer for *J. Comput. Phys.* and *J. Sci. Comput.* since 2025

## Honors and Awards

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| • New Lotus Award, the 2023 SGY Rose Scholarship                                                                                                                                           | 06/2024 |
| • USTC Outstanding Undergraduate Award                                                                                                                                                     | 06/2023 |
| • “Chia-Chiao Lin” Gold Medal in Applied and Computational track & Team Silver Medal & Excellence Prize in Analysis and PDEs track, the 14th S.-T. Yau College Student Mathematics Contest | 06/2023 |
| • Gold Prize, USTC Outstanding Student Scholarship                                                                                                                                         | 10/2022 |
| • Excellence Prize in Analysis and PDEs track, the 13th S.-T. Yau College Student Mathematics Contest                                                                                      | 08/2022 |
| • China National Scholarship                                                                                                                                                               | 12/2021 |
| • Second Prize, the 13th Chinese Mathematics Competitions                                                                                                                                  | 12/2021 |
| • China National Scholarship                                                                                                                                                               | 12/2020 |
| • Third Prize, USTC Freshman Scholarship                                                                                                                                                   | 09/2019 |

## Professional Skills

- Programming: MATLAB, C++, Fortran, Python, MPI, OpenMP

- Software:  $\text{\LaTeX}$ , Mathematica, NGSolve, FEniCS, MFEM, Tecplot
- Language: Mandarin Chinese, English

## Extracurricular Activities

- USTC road cycling team member, USTC 09/2019–06/2023
- Monitor of class 2019-3 for math-majored students, SGY, USTC 03/2022–06/2023

last update: May 12, 2026